

Preliminary Empirical Results

Testing the Zoning–Transportation Mechanism

Ke Jiang

Southern Methodist University

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Overview: Three Proposition Tests

Empirical chain from the model's FOC

- ➊ **P1.** High employment concentration $\Rightarrow$ tighter residential zoning
- ➋ **P2.** Transit access raises population pressure $\Rightarrow$ tighter zoning response
- ➌ **P3.** Restrictive land use $\Rightarrow$ lower metropolitan transit investment

Data

- Tract-level housing supply elasticity $\hat{\gamma}_i$: Baum-Snow and Han (2024), Dallas metro (N = 1,016 tracts)
- Employment inflows L_i : LODES 2023, aggregated to Census tract
- Transit access: DART station locations (148 stations), 0.5-mile buffer
- Cross-metro zoning: WRLURI from Wharton; transit network: Transit Explorer (40 matched metros of 306)

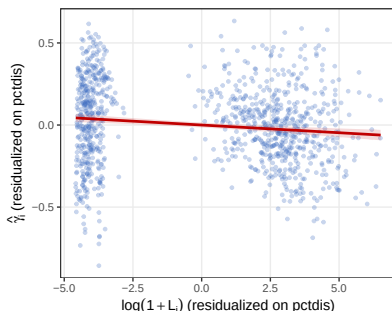
Tests are equilibrium consistency checks, not causal estimates. Each proposition maps to a model FOC.

P1: Employment Concentration and Zoning Restrictiveness

Prediction. From the municipal FOC,

$$\frac{\partial \hat{\gamma}_i}{\partial \log L_i} < 0.$$

High-employment tracts face larger marginal congestion costs from looser zoning.



Residualized on CBD distance ($\widehat{\text{pctdis}}$).

	(1) $\hat{\gamma}_{01a}$	(2) $\hat{\gamma}_{01a}$	(3) $\hat{\gamma}_{11a}$	(4) $\hat{\gamma}_{01a,sp}$
$\log(1 + L_i)$	-0.009*** (0.002)	-0.009*** (0.002)	-0.006*** (0.002)	-0.009*** (0.002)
pctdis_i		-0.087* (0.045)	-0.095** (0.045)	-0.040 (0.042)
R^2	0.016	0.020	0.011	0.017
N	1,016	1,016	1,016	1,016

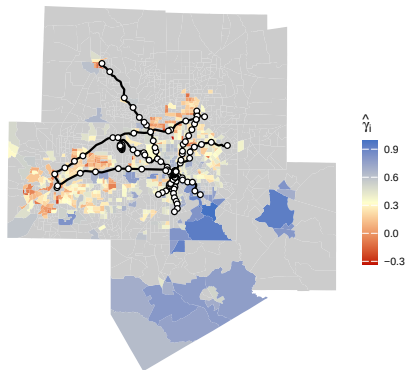
Result

Negative and significant across all four specifications. Consistent with the municipal congestion-cost FOC.

P2: DART Access and the Zoning-Tightening Response

Prediction. Transit access raises local attractiveness; municipalities respond by tightening residential capacity:

$$\hat{\gamma}_i|_{\text{near DART}} < \hat{\gamma}_i|_{\text{far}}.$$



	(1)	(2)	(3)	(4)
1[near DART]	-0.152** (0.059)	-0.205*** (0.058)	-0.178*** (0.057)	-0.177** (0.079)
pctdis		-0.412*** (0.059)	-0.384*** (0.059)	-0.384*** (0.059)
log(1 + L _i)			-0.032*** (0.007)	-0.032*** (0.007)
1[near] × pctdis				-0.011 (0.418)
R ²	0.011	0.087	0.118	0.118
N	581	581	581	581

N = 581 Baum-Snow tracts matched to 2020 Census

geometry; 15 have DART-adjacent centroid.

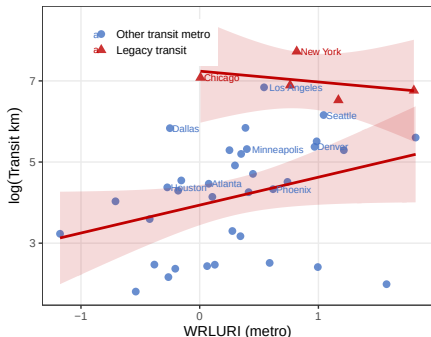
Result

Near-DART tracts have elasticities 0.18 units lower. Interaction insignificant (few suburban near-station tracts).

P3: Zoning Restrictiveness and Metro Transit Networks

Prediction. Restrictive zoning suppresses ridership density, lowering transit ROI:

$$\frac{\partial \log(\text{TransitKm}_m)}{\partial \text{WRLURI}_m} < 0.$$



40 US metros with existing transit matched to Baum-Snow; red = legacy systems.

	All metros ($N=306$)			Transit only	
	(1)	(2)	(3)	(4)	(5)
WRLURI	0.538*** (0.119)	0.193* (0.100)	0.196* (0.101)	0.965*** (0.284)	0.849** (0.332)
log(tracts)		0.847*** (0.068)	0.844*** (0.068)	0.995*** (0.208)	0.731** (0.271)
FracUnavail			-0.147 (0.431)		0.285 (1.413)
Legacy excl.?	—	—	—	No	Yes
R^2	0.063	0.383	0.383	0.484	0.274
N	306	306	306	40	35

Sign reversal — caution

WRLURI is *positive*: coastal legacy metros (NY, SF, Boston) are both more restrictive *and* have larger transit systems. This reflects historical endogeneity, not a refutation of the model.

Interpretation and Next Steps

What the tests show

- **P1** ✓ Employment-concentrated tracts are systematically less permissive, consistent with the municipal congestion-cost FOC.
- **P2** ✓ DART-adjacent tracts are significantly less permissive than matched non-adjacent tracts, consistent with the transit-tightening channel.
- **P3** — Cross-metro raw correlation is *positive*: legacy transit cities are restrictive and transit-rich. Historical endogeneity prevents direct inference.

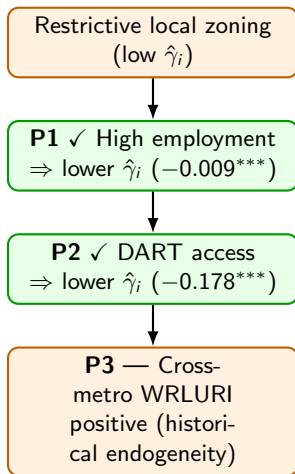
Identification strategy needed for P3

- Control for pre-automobile urban form (1920 density)
- Instrument WRLURI with state-level regulation history or terrain
- Use planned-but-unbuilt lines to isolate transit supply variation
- Restrict to metros without pre-war rail (cleaner counterfactual)

P2 refinements

- Expand to other metros in Baum-Snow sample
- Use distance to nearest station (continuous)
- Historical station-siting instrument for causal test

Summary



P1 and P2 provide within-DFW equilibrium support for the model's municipal FOC.
P3 requires causal identification via historical or geographic instruments.